

Syllabus (Tentative)

Fall Semester 2026 (Tentative)

Course Title	(Tentative) Special Topics in Chemistry (Computational Chemistry and Chemical Reactions) (화학특강(계산화학과 화학반응))	Course Code		To Be Determined		Credit	3	Classroom	To Be Determined
		Type		Elective		Hours	3		
Teacher	Name	Class No.	Office	Office Hour*		Contact		E-mail	
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* Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Description

Quantum mechanics are central tools to understand the fundamentals of modern chemistry. Computation chemistry has become an indispensable mean to perform quantum mechanical molecular modeling to deepen understanding of organic and inorganic compounds and their associated reactivities. The Computational Chemistry and Chemical Reaction course will cover density functional theory (DFT) analyses of the properties and reactivities of organic and organometallic compounds. The course will address molecules, intermediates, transition states and their associated chemical reactions in physical chemistry perspective. Various computational means will be delivered and practiced to evaluate the given chemical states and situations.

* Prerequisite courses: General Chemistry 1, General Chemistry 2, General Chemistry Laboratory 1, and General Chemistry Laboratory 2

2. Core Target Competencies

Subject Core Competencies						
Analytical and Creative Thinking	Complex Communication	Integrity and Ethical Decision Making	Complex Problem-solving Skills	Critical Thinking Skills	Creativity	Ability to Collaborate
The Computational Chemistry and Chemical Reaction course aims to cultivate the following competencies: Analytical and Creative Thinking / Complex Communication / Leadership and Teamwork / Global Perspective / Adaptability, Progression, and Spirit of Adventure / Integrity and Ethical Decision Making / Habits and Mindset / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate / Judgement and Decision-making Skills / Service Orientation / Negotiation Skills / Cognitive Flexibility						

3. Textbooks and References

3.1. Textbook

(1) ORCA 6.1 Manual from Max-Planck-Institut für Kohlenforschung (<https://www.faccts.de/docs/orca/6.1/manual/>)

- (2) ORCA 6.1 Tutorials (<https://www.faccts.de/docs/orca/6.1/tutorials/>)

3.2. References

- (1) Organic Chemistry, 9th Edition - by John McMurry (Cengage Learning)
- (2) Inorganic Chemistry, 5th Edition - by Gary L. Miessler, Paul J. Fischer, and Donald A. Tarr (Pearson)
- (3) Advanced Organic Chemistry, Part A: Structure and Mechanisms, 5th Edition - by Francis A. Carey and Richard J. Sundberg (Springer)
- (4) Pitfalls in Computational Modeling of Chemical Reactions and How To Avoid Them. Ryu, H.; Park, J.; Kim, H. K.; Park, J. Y.; Kim, S.-T.; Baik, M.-H. *Organometallics* **2018**, *37*, 3228-3239.
- (5) Other recent journal articles will be used as references for the course.

4. Class Method

4.1 Class Management

- (1) The course is conducted through (a) lectures and (b) student-centered learning activities. Students will be engaged with density functional theory (DFT) analysis programs and visualization means. (e.g., ORCA, Gaussian, Avogadro, GaussView, Chemcraft, and many others.)
- (2) Assessment will be performed based on both examination and non-examination criteria. (c) One mid-term exam will be conducted as part of the examination evaluation. (d) One final project, (e) assignments, (f) attitude, (g) in-class participation, and (h) attendance will be evaluated as part of the non-examination criteria.
- (3) (a) The lectures will be delivered in class using the board and/or presentation materials. Students are required to take handwritten notes in every class to enhance their summarization, clarification, and graphical description skills. The students' in-class notes will be submitted and assessed based on their content.
- (4) (b) Students will work in teams of two or three and will participate in team-based problem-solving activities during class. They will examine the learning materials by collaboratively performing the DFT quantum chemical analysis.
- (5) (b) The team activities will be evaluated on the individual's contribution basis. Students will participate in the team activities with acceptable rate of contribution and teamwork.
- (6) (c) A mid-term exam will be conducted to evaluate the students' analytical thinking, complex problem-solving skills, creativity, and critical thinking skills.
- (7) (d) A computational chemistry final project will be assigned to the students. Students will perform a DFT quantum chemical analysis to given chemical reactions or status. The evaluation will be based on the final presentation, report, and raw computational materials performed/submitted.
- (8) (e) An assignment will be given to the students after finishing each chapter and/or learning content. Students are required to address and submit the given problems to enhance their computational chemistry technics and complex problem-solving skills.
- (9) (f) Attitude will be evaluated to promote and encourage the learning environments. The evaluation on the attitude criterion will be fully depend and rely on the instructor(s)'s discretion.
- (10)(g) Students are required to participate in discussions regarding the given topics and questions to enhance their communication and critical thinking skills. The students' participation will be assessed based on their content and activity. The evaluation on the in-class participation criterion will be fully depend and rely on the instructor(s)'s discretion.
- (11)(h) Students are obligated to attend the class to learn the contents of the course. The evaluation on the attendance criterion will be fully depend and rely on the instructor(s)'s discretion and/or the Korea Science

Academy Regulations.

- (12) All of the students' evaluation materials (e.g., lecture notes, presentation files, reports, raw computational data, etc.) must be submitted to the Google Classroom.
- (13) Any late submission after the due date or time will not be accepted, resulting in 0 point for the related criteria.

4.2 Academic Integrity Policy

- (1) The diversity of the class members ought to be respected. Any type of discrimination regarding ethnicity, nationality, language, personal orientation, etc., will not be tolerated.
- (2) (EC) All communications will be conducted in English as part of addressing the language diversity of all members. Other languages are not allowed unless permission is granted by the instructor.
- (3) Plagiarism and copying are strictly prohibited. Citations and references must be provided on a sentence-by-sentence basis whenever a source is used. Any sequence of six or more consecutive words taken from another source will be regarded as plagiarism. Note that paraphrasing is required whenever sources are referred to. All visual and technical materials (e.g., figures, equations, graphs, tables, and code) must be properly cited, and any reuse must comply with applicable permissions and licensing. Students must comply with all other regulations regarding academic integrity.
- (4) Using AI (artificial intelligence) tools (e.g., ChatGPT) for academic support is permitted; however, students must verify and study the original sources cited or referenced. In addition, students must cross-check information from multiple scholarly references (e.g., peer-reviewed books or academic journals). In this course, AI-generated content is not accepted as a citable source or reference. Submitting AI-generated content without proper citation to original sources will be regarded as plagiarism and unauthorized copying.
- (5) The use of AI tools (e.g., ChatGPT) to write or generate the contents of submitting materials is strictly prohibited. The school may use plagiarism-detection tools (e.g., CopyKiller, Turnitin, etc.).
- (6) If a submitted content exceeds a specified similarity threshold (e.g., 15% or higher) or is detected as AI-generated contents by CopyKiller mechanism (<https://ksa.copykiller.com/>) in portion or all, it will be considered plagiarism/unauthorized copying regardless of student(s)'s opinion or argument.
- (7) Students are responsible and obligated to check any possibility for plagiarism. All of the materials and contents must be verified by CopyKiller (<https://ksa.copykiller.com/>) by individual students prior to the final submission. The detail plagiarism analysis report on the submitting contents or materials must be provided within.
- (8) Any failure in complying the section [4.2 Academic Integrity Policy] or general academic integrity will result in an FAIL grade for the course. Furthermore, the student will receive official disciplinary punishment according to the Korea Science Academy Regulations.
- (9) Any type of misconduct (e.g., cheating, copying, possession of electronic devices during exam, refusing to follow instructions, dishonorable actions, other unauthorized activities not related to the course, etc.) will automatically result in an FAIL grade for the course. Furthermore, the student will receive official disciplinary punishment according to the Korea Science Academy Regulations.

5. Evaluation

5.1 Evaluation Criteria

Mid-term Exam	Final Project	Lecture Note	Assignments	Attitude	In-class Participation	Attendance	Sum
25	25	10	15	10	10	5	100

5.2 Evaluation Method

5.2.1 Mid-term Exam (25%)

- (1) One mid-term exam will be held after covering specific chapters and/or learning contents. Understanding basic knowledge on quantum mechanics, computational chemistry, reaction mechanisms and many others will be covered in the mid-term exam. Creative thinking, cognitive flexibility, literal explanation, and graphical demonstration will be important competencies for the evaluation.
- (2) The mid-term exam will be held for approximately 180 minutes. The exam duration may be adjusted based on the given circumstances.
- (3) Each student is allowed to prepare 3 pages of Memory Sheets for the exam. The blank Memory Sheets will be provided to the students at the start of the course. The prepared Memory Sheets may be possessed by the student during the exam only if (a) they are approved by the instructor on the exam date, (b) they include the student's handwritten Honor Code, and (c) they bear the signatures of both the student and the instructor.
- (4) Along with the other criteria, the exam will be administered in strict accordance with the Korea Science Academy Regulations and the [4.2 Academic Integrity Policy] section.
- (5) The make-up score for an mid-term exam missed due to an excused absence or a medical absence shall be calculated as follows. The baseline score and percentile will be determined by the Final Project criterion. The make-up score for missed mid-term exam will be determined by multiplying the baseline percentile by 100% (excused absence) or 80% (medical absence). Further details are available in the Korea Science Academy Regulations.

5.2.2 Final Project (25%)

- (1) Each team of the two or three students will perform computational analysis of organic/organometallic chemical phenomena. The theoretical background and program utilization will be covered in class.
- (2) Each team is required to discuss the topic and detailed computational methodology with the instructor before the project begins.
- (3) Each student will present their project results in class and submit the final report. The evaluation will be performed by the submitted contents and interviews involving questions and answers.
- (4) The total percentile for the Final Project is 25%. The detailed assessment criteria are:
 - (a) setting up a creative and complex topic (5%)
 - (b) computational method and progress (3%)
 - (c) result analysis and discussion (5%)
 - (d) presentation skills (3%)
 - (e) final report content (9%)
- (5) The project results will be summarized and submitted as a PDF report. The raw computational data will be submitted as text files. The content of submitted PDF files must be clear and easy to read. Any vague content in terms of light and dark contrast, color, etc., will not be assessed, resulting in 0 point for the related criteria. The use of a scanner is recommended over the use of a cell phone camera if necessary.

5.2.3 Lecture Note (10%)

- (1) Students are required to take handwritten notes in every class. Each set of notes must be submitted as a single PDF file. Taking handwriting notes on tablets (e.g., iPad, Galaxy Tab, etc.) is allowed. However, lecture notes created using word processing software (e.g., Microsoft Word, Pages, Hangeul, etc.) are not allowed unless permission is granted by the instructor.

- (2) The graphical figures must be drawn by hand. This policy is intended to enhance students' summarization and graphical description skills.
- (3) The content of submitted PDF files must be clear and easy to read. Any vague content in terms of light and dark contrast, color, etc., will not be assessed, resulting in 0 point for the related criteria. The use of a scanner is recommended over the use of a cell phone camera.

5.2.4 Assignments (15%)

- (1) Students will receive assignments after finishing each chapter and/or learning content. The assignment may involve questions on the computational chemistry methodologies, quantum mechanics, chemical reaction mechanisms, and many others.
- (2) Individual student is required to submit the handwritten answers for the assignments. Each set of the answers must be submitted as a single PDF file. Taking handwriting answers on tablets (e.g., iPad, Galaxy Tab, etc.) is allowed. However, answer sheets created using word processing software (e.g., Microsoft Word, Pages, Hangeul, etc.) are not allowed unless permission is granted by the instructor. The raw computational data will be submitted as text files.
- (3) The graphical figures must be drawn by hand. This policy is intended to enhance students' summarization and graphical description skills.
- (4) The content of submitted PDF files must be clear and easy to read. Any vague content in terms of light and dark contrast, color, etc., will not be assessed, resulting in 0 point for the related criteria. The use of a scanner is recommended over the use of a cell phone camera.

5.2.5 Attitude (10%)

- (1) Students will start with 0% in the Attitude criterion. Students may earn points through positive behaviors and actions, while negative actions and behaviors will result in point deductions.
- (2) Attitude will be evaluated as follows: Positive contributions to the learning environment may be rewarded with +0.5%. Behaviors that disrupt the learning environment will be penalized -0.5% per occurrence (e.g., unauthorized use of a non-English language, studying for other classes, unauthorized chatting, taking a nap, etc.). Unauthorized use of cell phones and computers will result in a -2% deduction per occurrence. Unauthorized consumption of food and beverages is not permitted in the classroom.
- (3) The assessment of the Attitude criterion will be subject to the instructor's discretion.

5.2.5 In-class Participation (10%)

- (1) Students will start with 0% in the In-class Participation criterion. Students may earn points through positive behaviors and actions, while negative actions and behaviors will result in point deductions.
- (2) In-class participation will be evaluated as follow: Active participation and communication during class may result in earning positive points. Low participation and weak involvement in discussion will lead to the point deductions.
- (3) The assessment of the In-class Participation criterion will be subject to the instructor's discretion.

5.2.6 Attendance (5%)

- (1) Attendance will be recorded and graded as follow: Students will receive 5% for no unexcused absence and no late arrival. Being late for class will be penalized by -0.5% per each period. One period of unexcused absence will be penalized by 1%. Two periods of unexcused absences will be penalized by 3%. Three periods of unexcused absences will be penalized by 5% and will not receive any percentile in Attendance criterion. Four

periods or more unexcused absences will be result in FAIL grade of the course.

6. Criteria for Awarding Credits

A (100.00 ~ 75.00)			B (74.99 ~ 60.00)			C (59.99 ~ 45.00)			D (44.99 ~ 30.00)			F
A+	A0	A-	B+	B0	B-	C+	C0	C-	D+	D0	D-	
100.00	89.99	79.99	74.99	69.99	64.99	59.99	54.99	49.99	44.99	39.99	34.99	29.99
~	~	~	~	~	~	~	~	~	~	~	~	~
90.00	80.00	75.00	70.00	65.00	60.00	55.00	50.00	45.00	40.00	35.00	30.00	0

- (1) The course will follow an absolute grading (criterion-referenced grading) system. Grades will be assigned based on the over evaluation percentile that each student achieves.
- (2) Students with outstanding academic achievement may be recommended for the Academic Excellence Award and the Principal's Award, and their performance is recorded under the core competencies in the Subject Details. However, students with poor academic attitude, in-class participation, or attendance will be excluded from consideration regardless of the achievement.

- Attendance for at least three-quarters of the total class hours is required to earn credit. An excessive number of excused or medical absences may result in a FAIL grade for the course, in accordance with the Korea Science Academy Regulations.
- For PASS/FAIL courses, if the instructor determines that a specific evaluation component is insufficient, an additional assessment may be conducted.
- The method for calculating the baseline score used to award credit to students with absences will be determined by the committee within the Department of Chemistry and Biology at the Korea Science Academy.
- If the baseline score cannot be calculated, the grading method will be determined by the committee within the Department of Chemistry and Biology at the Korea Science Academy.

7. Weekly Plans

Week (period)	Topic and Contents	Notes
Week 1	Introduction to Computational Chemistry and Quantum Mechanics	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 2	Frontier Molecular Orbital (FMO) Theory	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 3	Natural Bond Orbital Analysis	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision	

	Making	
Week 4	Geometry Optimization and Vibrational Mode Analysis of Three-Dimensional Molecules and Intermediates	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 5	Construction and Evaluation of Three-Dimensional Transition States: Intrinsic Reaction Coordinate (IRC) Analysis	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 6	Potential Energy Surface (PES) Scan Method	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 7	Visualization of Orbital Interactions: Intrinsic Bond Orbitals	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 8	Midterm Exams	

Week 9	Solvation Model: PCM and SMD Methods	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 10	Evaluation of Gibbs Free Energy Profiles (1): Reaction Kinetics and Thermodynamics	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 11	Evaluation of Gibbs Free Energy Profiles (2): Consideration of Solvation Models	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-	

	solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 12	Prediction of Nuclear Magnetic Resonance (NMR) Spectra	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 13	Prediction of UV-vis Spectra: Time-Dependent Density Functional Theory (TD-DFT) Analysis	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 14	Final Project Preparation Week	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 15	Final Project Presentation Week	
Core Competencies	Analytical and Scientific Thinking / Ability to Explore / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability to Collaborate/ Subjective and Complex Communication / Integrity and Ethical Decision Making	
Week 16	Make-up class week	-
Week 17	Final Exams	-

8. Note

The course plan and policies may change at the instructor's consideration, depending on the given circumstances.